

Education

B.S. Computer Science (Turing Scholar), B.S. Mathematics

GPA: 4.0

University of Texas at Austin

Expected: May 2028

- Relevant Enrollments: Algebraic Structures, Operating Systems Honors, Real Analysis, Computer Architecture Honors, Vector Calculus Honors, Data Structures, Discrete Math, Linear Algebra, Graph Theory, Number Theory, Principles of ML
- Other Enrollments: Genetics, Evolution, Biochemistry, Microbiology, Organic Chemistry

Experience

Directed Reading Program | *Summer Research*

Jun 2026 - Aug 2026

- Helped develop normalizing-flow architecture for PDFs on quotient manifolds
- Implemented the genus- g application including $\mathbb{H}^2/\Gamma \rightarrow \mathbb{R}^2$ maps and tilings of $\mathbb{H}^2$
- Trained model on hyperbolic heat kernel and verified via semigroup property $< 10\%$ rel. err.
- Achieved $KL < 0.03$ for best distribution (Gaussian) and $KL < 0.64$ for worst distribution
- Implemented Riemannian Flow Matching as a baseline to verify our model's competitiveness
- Co-authored a paper submitted to NeurIPS; contributed the genus- g section as second author
- Tech and Skills: Topology, Quotient Manifolds, PyTorch, NumPy, LaTeX

Applied Research Laboratories | *Apprenticeship Program*

Jun 2025 - Aug 2025

- Came up with and optimized a GPU-based algorithm for signal acquisition
- Achieved 20x speedup on the GPS L1 C/A band using a 2019 GPU
- Extracted partial ephemeris data bits which proves real-time acquisition
- Developed a 50 FPS display of satellite heatmaps across time and Doppler using OpenGL
- Tech and Skills: PyTorch, NumPy, CUDA, C++, FFTs, math of signal processing

BioBQ | *Data Analyst Intern (remote)*

Aug 2024 - May 2025

- Trained and deployed a YOLO v8 model to analyze cell growth and type
- Prototyped an approach for bovine cell monitoring using transducers and the Doppler effect
- Tech and Skills: Pandas, Keras, ML knowledge

Institute for Computing in Research | *Research Intern*

Jul 2024 - Aug 2024

- Deployed a R-based model that found TCs (transmission clusters) from phylogenetic trees
- Analyzed state-by-state SARS-CoV-2 TCs to find trends and verify model's accuracy
- Conducted a literature review and wrote a report presented at TyRex
- Tech and Skills: R, LaTeX (+Beamer), Pandas, statistics, phylogenetic methods

Projects

ViralSim | *C++, Bash, Python, Snakemake*

- Built agent-based pandemic simulator with real-time phylogeny generation
- Designed a language with conditional logic, functions, and simulation step options
- Implemented the corresponding interpreter to allow for user-specified config files
- Used simulated DNA data to evaluate phylogenetic methods
- Presented at the Oden Institute at UT as part of a student event

Skills

Proficient: C, Python, C++, Verilog, R, bash, Java, Git, LaTeX, NextStrain, Unix

Familiar: Lua, Ngspice, CUDA, Snakemake